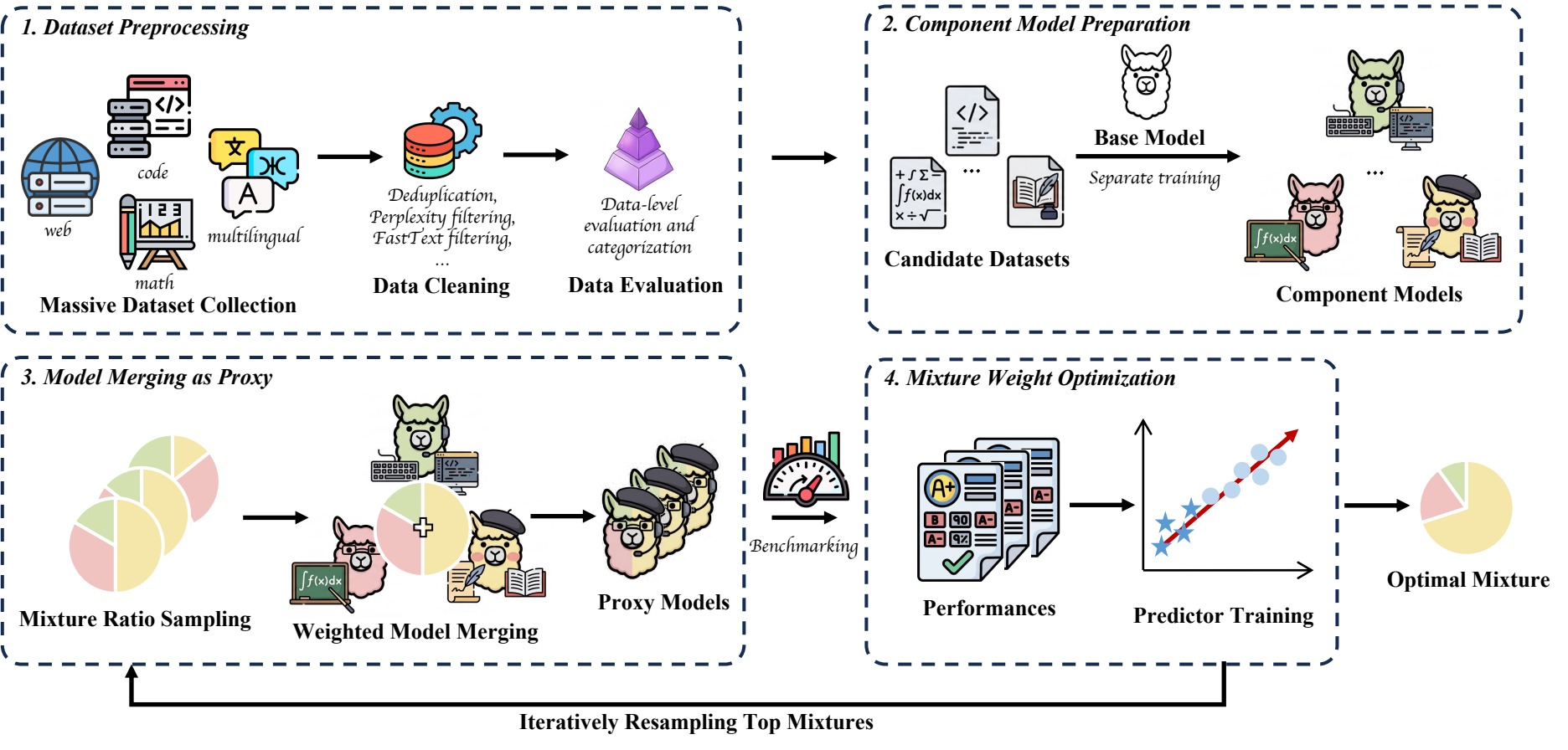
1. Dataset Preprocessing
web
code
math
multilingual
Massive Dataset Collection
Deduplication,
Perplexity filtering,
FastText filtering,
...
Data Cleaning
Data-level
evaluation and
categorization
Data Evaluation
2. Component Model Preparation
Candidate Datasets
Base Model
Separate training
Component Models
3. Model Merging as Proxy
Mixture Ratio Sampling
Weighted Model Merging
Proxy Models
Benchmarking
4. Mixture Weight Optimization
Performances
Predictor Training
Optimal Mixture
Iteratively Resampling Top Mixtures